

# Coursework Specification: The Invisible Caregiver

## Project Goal

Design and implement a context-aware assisted living system that uses Small Language Models (SLMs/LLMs) to monitor an elderly resident's safety and daily routines without violating privacy via cameras.

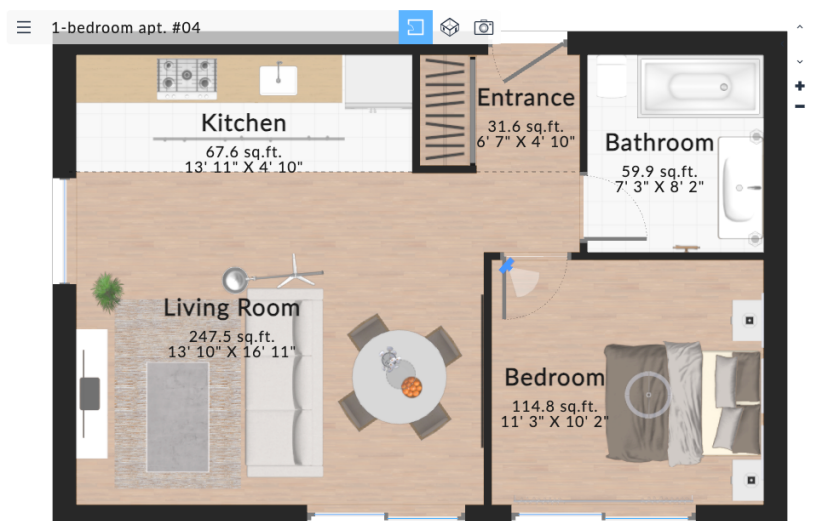
The system implements a complete smart home data acquisition, storage and processing pipeline, leveraging edge computing technologies and AI. The system ingests data from sensors, manages their storage and semantic organisation, and exposes the data to third-party applications and external users as required.

The elderly person is in good physical health, but must take some medication regularly. As most elderly persons, the home occupant may be a little forgetful, showing mild signs of dementia. The system must be able to support their basic activities and routines, such as cooking, washing themselves, engaging in some social or leisure activities, allowing them to live well independently.

The system collects data from the sensors and stores them in a small home server. The data is accessible to an edge AI device (Jetson Nano) that is able to analyse it and produce a natural language response, summarising arising issues, or answering questions of remote care-takers about the home occupant's wellbeing, through a chat interface.

## Part 1: System Design & Sensor Mapping

Students must design a "Digital Twin" of a single-resident apartment. The apartment consists of 1 bedroom, 1 kitchen, 1 living room and 1 bathroom. You can design your own floorplan, but some ideas can be found here: <https://www.spacedesigner3d.com/gallery/showcase/simple-one-bedroom-apartment-small-spaces>



- **Sensor Selection:** Define at least **6 distinct sensor types** (e.g., PIR motion, bed pressure mats, contact sensors for fridge/cabinets, power monitoring plugs, water flow meters, vibration sensors). These sensors will be placed around the home and you can install as many of each type of sensors as you want. For the sake of privacy and security, no cameras or microphones are allowed.
- **Placement Strategy:** Justify the location of each sensor. *Why place a vibration sensor on the medicine cabinet versus a contact sensor?*
- **Data Schema:** Define a consistent JSON schema for the data packets.

## Part 2: Data Generation & Simulation

Students must write a **simulation** script (Python/Node.js) that generates **three distinct scenarios** of sensor data of increasing levels of danger. For example:

1. **The "Normal Day":** A consistent routine of waking, eating, and hygiene.
2. **The "Subtle Decline":** A pattern showing forgetfulness (e.g., fridge left open, sedentary behavior for 6+ hours).
3. **The "Acute Hazard":** A dangerous event (e.g., stove left on while the resident is in the bedroom, or a 3:00 AM exit from the front door).

**Technical Requirement:** Data must be published to an IoT Platform (e.g., **ThingsBoard**) via **MQTT** or **REST API**. Select an open-source platform that is locally installable and justify your choice.

## Part 3: Semantic Processing on Jetson Nano

The Jetson Nano will act as the "Brain." It must pull data from the IoT platform and process it using a local LLM (e.g., Llama-3-8B or Phi-3.5 via llama.cpp).

### Requirement A: The Safety Auditor (Automated)

- **Frequency:** Runs as a cron-job or loop every 60 minutes.
- **Input:** The last 1 hour of sensor data + a "Care Plan" (Pre-defined rules in the prompt).
  - The care plan may include instructions about medicine requirements etc.
  - Standard information like "hands must be washed after every cooking activity", "going to the toilet must happen at least once a day" etc. can also be included.
- **Output:** If a discrepancy is found, generate a **Structured Alert** (JSON) for a dashboard or SMS.

### Requirement B: The Narrator (On-Demand)

- **Function:** A conversational interface where a caretaker can ask natural language questions.
- **Input:** The last 24 hours of data.
- **Prompting:** The student must craft a "System Prompt" that ensures the LLM provides **empathetic yet factual** summaries.
  - **Example Query 1:** "Did Mom have a healthy morning?"
  - **Example Query 2:** "How was Mom's day today?"
  - **Example Query 3:** "Did my Mom forget to take her medication today?"

## AI Analysis Example

### System Prompt:

*You are a "Compassionate Context Engine" for an elderly care facility. Your job is to translate raw sensor logs into a warm, natural language summary for a family member. Do not list raw timestamps unless necessary. Focus on "human" activities like waking up, hygiene, eating, and safety.*

### User Prompt:

*Please analyze Mary's routine and provide a brief summary. Specifically address the following aspects:*

- 1. Did she wake up at a normal time?*
- 2. Is there evidence she took her medication and had breakfast?*
- 3. Did she leave the house?*

[INSERT JSON DATA HERE]

### Example Response:

"Mary had a steady start to her morning! She was up and out of bed by 7:05 AM. She spent some time in the bathroom (likely a shower) before heading to the kitchen. It looks like she had a hot drink around 7:40 AM and, most importantly, the sensors show she accessed her medication box at 8:00 AM. After relaxing in her favorite chair for an hour, she stepped out of the front door at 9:20 AM—likely for her daily morning walk."

## Project Deliverables and Proposed Timeline

| Proposed Deadline | Deliverable                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8 May 2026        | Project Analysis and Design Report        | Your initial design for the smart home, including sensor type definitions, sensor placements, sensor data schemas, IoT management platform choice and justification, sensor modelling in IoT platform. Include an introduction, related work section and design section as well as references. Use the <a href="#">ACM manuscript template</a> for your document. Document your work with diagrams, screenshots or any other materials as you think are required (figures and tables). |
| 29 May 2026       | Data simulator scripts and revised report | Python code to generate time series from the sensors and send them to the IoT platform for storage. Extend the initial report to include details about how you simulated the 3 scenario danger levels and explain the logic in your implementation. Document your work with diagrams, screenshots or any other materials as you think are required (figures and tables).                                                                                                               |
| 26 June 2026      | AI implementation and final report        | Python code to extract data from the IoT platform, pass them to the Jetson Nano and process them. Extend the initial report to include your strategies for prompt engineering, final selected prompts, and assessment of the results (e.g. did the LLM capture all the important details? Did it miss any dangerous or critical events?) Document your work with diagrams, screenshots or any                                                                                          |

|              |                     |                                                                                                                                                            |
|--------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                     | other materials as you think are required (figures and tables).                                                                                            |
| 26 June 2026 | Demonstration video | Document your system in a demonstration video, showing the architecture, principles and final outputs from interaction with the AI system (5 minutes max). |

- This is a TEAM coursework – up to 3 students per team.
- Submit your FINAL deliverables (final report, code and video) on e-class ONLY.